

## Problem Sets After Test 2

Only turn in problems that are **not** bracketed. Bracketed problems are additional problems you can look at. Round brackets indicate problems that may help you with problems that are assigned; square brackets are additional problems on material that you should know, but you are not required to write up solutions; curly brackets are truly optional and may contain extra nuggets that you will not be required to know but may be interested in.

Additional assignments will be filled in over time.

| notation    | meaning                                                                       |
|-------------|-------------------------------------------------------------------------------|
| unbracketed | assigned problem – turn these in for grading                                  |
| ()          | helper/warm-up problem                                                        |
| []          | additional problems (you are responsible for content, but don't turn them in) |
| { }         | covers optional material                                                      |

| Due      | Task  | Source                        | Problems                                                                                                                                                                                                                                                             |
|----------|-------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wed 4/7  | PS 13 | Rosen 13.3                    | (11) testing strings <b>12</b> testing strings <b>16–17</b> what language? <b>25–26</b> create DFA                                                                                                                                                                   |
| Fri 4/9  | PS 14 | Rosen 13.3                    | <b>19–20</b> what language? [32] create DFA <b>34</b> create DFA <b>36</b> create NFA <b>45–46</b> what language? [47] what language? <b>55–56</b> create DFA, NFA<br><b>Note: Please do 55 and 56 in this order: 55a, 56a, 55b, 56b, 55c, 56c.</b>                  |
| Wed 4/14 | PS 15 | Rosen 13.3<br>Rosen 13.4      | <b>52–54</b> NFA to DFA<br>[1] <b>2abde</b> describe in words [3] <b>4</b> does it belong? <b>6</b> regex [7] regex                                                                                                                                                  |
| Mon 4/19 | PS 16 | Rosen 13.4                    | <b>14abc</b> grammar to NFA <b>15–17</b> NFA to grammar                                                                                                                                                                                                              |
| Mon 4/21 | PS 17 | Rosen 13.4<br>DFA/NFA → regex | <b>12ab</b> regex to NFA <b>13a</b> regex to NFA<br>Convert the following DFAs or NFAs into equivalent regular expressions using our algorithm for doing so. Show the intermediate steps: <b>13.3.43 13.3.45</b> (Note: these are not the instructions in the book!) |